

largest is missing

Rethink a bit the focus

Difficult to read the result $\leftarrow$ data does not allow for that

see ① and ②

TO STOP OR NOT TO STOP: A RESEARCH ON WHETHER MOL BUBI USERS SWITCH BIKES TO SAVE MONEY

Conductors of the Research	Fábián Máté Garas, Botond Rudolf Ujvári
Research question	<i>Do the users of the MOL Bubi bikes exploit the pricing structure of the 30 minutes of free bike rides? What measure of tradeoff do they use between the extra cost and the extra time it takes to change bikes?</i>
Methodology	Tracking bike users, to see the free ride time abuse, using logistic regressions and visualizations.
Research result	Optimize the pricing structure of the shared bicycle system by finding the behavioral patterns of the users.
Extensions and Possible future research topics	Price elasticity of the MOL Bubi shared bicycle system, the changes in the bike switching phenomenon in different times of the day.

① How many of such switches exist?
② Do we see a change in pricing?

RESEARCH PROBLEM / MOTIVATION

The MOL Bubi system was first initialized in 2014 in the capital city of Budapest. The main purpose of this bike sharing system was to create an alternative to public transport and cars which would make urban transport more efficient, while also helping to reduce harmful gas emissions in the process. During the first few years of the project, it proved to be a huge success. In the first year alone, between 2014 September and 2015 October, around 1 million bikes were rented out and used (BKK, 2015). The second year showed even more success in their numbers with a staggering 1.4 million rentals and over 2 million kms cycled throughout the city with MOL Bubi.

* Bubi fees are very low. Does this artificial pricing has a price elasticity?

Since September of 2014, the system of these shared bicycles was refined and improved to make renting more convenient, while also constantly updating the number of bicycles usable in the city of Budapest. The project, however, had a considerable fall in interest after the first few years due to several reasons, with assumptions of it failing overall. But in April 2021, several changes were implemented, such as lighter and much more agile new bicycles, and a more user friendly and reworked interface for the users to operate. This of course had a cost of having less bicycles in the city but presented a much more popular mode of transportation.

With the slow approach of the end of MOL Bubi 2.0, and the starting of a new era in the history of this service with a vastly extended number of bikes and the introduction of e-bikes, it is time to analyze its mechanics and inner workings of those who use the bike sharing system.

For those who are not frequent users of these bicycles, there are numerous amounts of stations laid out throughout the city of Budapest, where you can deposit and take these bikes. There are two ways to access them: You can either pay a per minute fee for the duration of time between picking up the bike from one station, and leaving it at another one, or you can purchase a pass for it, which grants you unlimited number of rides for a given period of time. However, in order to incentivize people to deposit the bikes back at the station rather than leaving them on the side of the road, only the bike rides that take less than 30 minutes are free for those who use passes to commute with the bicycles. After 30 minutes, additional per minute fees also have to be paid.

When a network of shared transportation, such as this, is created one of the most important tasks is to keep analyzing how it functions throughout the years. However, it is even more important to understand and utilize both the traveling habits of the users, (in order to be able to maximize the efficiency of the network that is created via the rides between the station) and the behavioral patterns which are reactions to the pricing structure for example. This way, new users can be drawn in, while keeping the old ones satisfied.

LITERATURE REVIEW

Bike sharing systems have been analyzed in several different aspects, and since we can find many of these globally in larger cities, it is a part of many urban daily commutes. Since it is a

2) shall we reduce the time to 15 min?
or increase to 45 min?

major part of transportation systems, its history and advancements have also been studied frequently (DeMaio, 2009).

One of the main aspects to look at is the allocation of the stations according to the geographical properties of the city (Zhang et al., 2018). Several studies analyze the population which utilizes these shared bicycle systems and the various inequalities within the population of users (Ogilvie and Goodman, 2012). But of course, using these bikes daily can also have many external effects, for example the reduction of harmful gas emissions (Chen et al., 2022) the health-related changes of the increased amount of bicycle rides (and including the danger factors) of city cycling (Woodcock et al., 2014). As the phenomenon of shared bicycle systems are appearing and being researched worldwide, the Hungarian counterparts are also frequently studied through domestic papers (Kovács and Gede, 2020) from several intriguing points of views, while also showing up in multiple international papers as well.

RESEARCH QUESTION

The question is, how do people use these bikes? Do they try to be as cost efficient during their rides as possible? Do they put the emphasis on taking scenic routes without prioritizing financial motivations? Or do they rather use these bikes to get to their destination as soon as possible, because they value their time more than extra costs?

These are the questions that interested us the most about this bicycle sharing system, since a major part of the mechanics of a service like this strongly depend on the behavior of its users. With a thorough analysis of the reactions to this type of pricing, both the optimization of the pricing structure is made possible, so that these user's preferences can be integrated into the further developments of MOL Bubi, while also uncovering the patterns of the consumer's behavior.

To summarize what our research is truly about, we propose the following research questions we aim to answer:

Do the users of the MOL Bubi bikes exploit the pricing structure of the 30 minutes of free bike rides? What measure of tradeoff do they use between the extra cost and the extra time it takes to change bikes?

And our second research question:

Is bike switching driven by rational behavior?

To answer this, we estimate how much they saved by switching bikes and compare it to the time they lost by making an additional stop and switching bikes.

This is interesting but lacks motivations!

DATA

Our data was sourced directly from BKK and broken down into the duration of months; it documents several useful aspects of every single MOL Bubi bicycle ride taken in the past years: Every ride is provided with a unique identifier number. For each of these journeys, we can observe the exact date and time the bike was rented out and it was started to be used, as well as the time when it was put back into one of the stations. Along with these two variables, the duration of every trip was also calculated and included in the measure of seconds. In order to be able to pinpoint the geographical starts and finishes for each journey, both the starting and ending stations were included in the data with several variables: These included the names of the stations, a special numerical identifier for each terminal and their exact coordinates provided in longitudinal and latitudinal variables. As each bike also has a unique id number, the tracking of which exact bike was used is also possible for us, thanks to the precision and accuracy of this dataset.

For our research however, the two most important parts of the data were the consumer id and the payment method for the rides. The payment method is an essential specifier in the data, since it shows that a pass was used to rent a bicycle, which pass is what makes the renters eligible for the first 30 minutes of the ride being free, since we are primarily interested in the use of this mechanic. Moreover, consumers who buy the monthly or yearly passes tend to be frequent users of these bicycles, giving us even more data to work with to compare what kind of routes they take during these rides.

For the designated consumer id, it will also play a vital part in our research, since it shows if the same customer swapped their bikes for a new one at one of the stations to avoid the additional costs during their travels, which gives us the possibility to track individuals who have swapped their bikes in the middle of their journey in order to avoid the extra cost that would come with the over half an hour ride.

Not clear from the data, how you can measure price elasticity.

What we would really like to find are clear behavioral patterns indicating that many MOL Bubi pass users consciously optimize their rides to avoid additional fees after the 30-minute free period.

I expect a small fraction of such users.

Specifically, we would anticipate a noticeable concentration of ride durations just below the 30-minute threshold and frequent instances of users ending and restarting rides within a short time window, often at or near the same station. Furthermore, we expect time differences when users during the morning hours are more focused on efficiency and evening users showing slightly higher flexibility and potential cost-saving behavior.

However, many challenges can arise during the analysis. First, the dataset may not capture the exact intentions of users, meaning that identifying cost-avoidance purely from ride patterns can only be inferred, not confirmed. Additionally, variations in weather conditions, traffic, or other seasonal effects may influence riding behavior and complicate comparisons across time periods. Another potential limitation is the lack of detailed demographic or socioeconomic data, which prevents the identification of user-specific characteristics that could explain different behavioral patterns.

Impact

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METHODOLOGY

We plan to search for and identify customers that have taken a bike ride to a certain point, deposited it within the 30-minute timeframe, then taken another bike out again to save the additional costs they would have received, if they had simply continued riding from their starting point to their intended destination.

After the data is filtered to the relevant subset, we will isolate rides made with active passes, since these are subject to the 30-minute free usage period. Within this dataset, we will examine whether users tend to end and restart rides shortly before or after the 30-minute limit in order to avoid extra charges. This will be done by tracking consecutive rides made by the same user and analyzing cases where a bicycle is returned within 30 minutes and another one is taken out from the same or a nearby station within a short time interval. This can be done, since we have unique identifiers for each user, so we can see if the same person switched their bike at the station.

We will compare the number and timing of such “bike swap” events across different periods of the day (e.g., morning versus evening commutes) and analyze the distribution of ride durations to detect clustering around the 30-minute threshold. This will determine the extent to which users may be adapting their cycling routes and routines specifically to take advantage of the free time slot.

We will also use statistical tests such as logistic regressions to determine whether these cost-avoidance behaviors differ systematically across user groups, time periods, or station locations. In addition, we will visualize these patterns using temporal and spatial plots to highlight hotspots of frequent bike-swapping behavior and identify broader trends across the city. These combined approaches will provide both a detailed quantitative viewpoint and an overview of how pass users respond to MOL Bubi’s pricing structure.

EXPECTED OUTCOMES

* How do you capture user groups?

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